EYAD JAWAD

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EDUCATION

University of Technology (UOT) Baghdad, Iraq

B.Sc. in Computer Science (Year 2) *Expected 2028*

- Achieved Class Valedictorian honors during the first academic year for ranking 1st in the cohort.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS, SQL

Core Concepts: Data Structures, Algorithms, Bit Manipulation, Image Processing, Asynchronous Programming, 3D Rendering, Backtracking, Software Testing, Documentation

Tools & Frameworks: Git, SQLite, WebGL, Pygame, BeautifulSoup, GTest, PyTest, Node.js/npm, Makefiles

KEY PROJECTS

Maze Game (C++, JavaScript, WebGL) GitHub Link

- Engineered highly optimized C++ maze generation and solving algorithms from scratch, completing generation in under 1ms (down from 517ms baseline) and solving via backtracking in under 15ms (down from 5626ms baseline), representing a 500× performance improvement.
- Developed a zero-dependency, interactive 3D rendering environment using raw WebGL and native JavaScript.
- Deployed production web build via GitHub Pages for public real-time browser-based testing.

Image Processing Engine & Technical Research (C++) GitHub Link

- Programmed a low-level computer vision pipeline executing Sobel filtering, Gaussian blur, and full Canny edge detection from scratch, incorporating multi-stage optimizations like non-maximum suppression (NMS) and hysteresis tracking.
- Built a custom 2D convolution matrix execution layer supporting dynamic kernel generation and direct binary parsing of raw BMP files.
- Authored a comprehensive technical research paper evaluating low-level kernel convolutions across 10+ academic sources.

Telegram Archiver (Python, Telethon) GitHub Link

- Developed asynchronous streaming data pipelines utilizing Telethon to index and archive over 100,000 chat messages alongside rich media attachments.
- Implemented a fault-tolerant checkpoint system providing precise execution state persistence and graceful connection error recovery.
- Designed robust structured relational and CSV schemas ensuring valid data integrity for multi-format parsing.

Huffman Compression Tool (C++) GitHub Link

- Coded an end-to-end compression utility from the ground up implementing custom Huffman encoding and decoding algorithms.
- Designed a proprietary binary archive file format (`.HUF`) featuring strict bit-level data packing to eliminate structural overhead.
- Optimized recursive binary tree traversal routines for deterministic serialized tree structure packing and unpacking.

ACHIEVEMENTS & LEADERSHIP

Winner – Excellence Hackathon 2026 University of Baghdad

- Secured 1st place at the College of Excellence software hackathon, out-competing a highly selective group of top-ranked first-year university candidates.

3rd Place – National Data Science Hackathon 2026 University of Al-Farabi

- Co-led a 3-person team through a high-pressure 24-hour national hackathon, placing 3rd out of over 30 competing teams comprising university seniors, advanced students, and graduates.

ADDITIONAL INFORMATION

- **Rapid Technical Progression:** High-velocity self-taught engineering trajectory with a proven capacity to independently research, architect, and optimize low-level systems.
- **Certifications:** Completed Harvard University's **CS50** (Computer Science Fundamentals), establishing deep baselines in rigorous algorithmic analysis and memory management.